

Module -III

Enzymes and Catalysis



R. K. Gothwal, PhD
Department of Bioengineering and Biotechnology
BIT Mesra, Ranchi
Jaipur-Campus

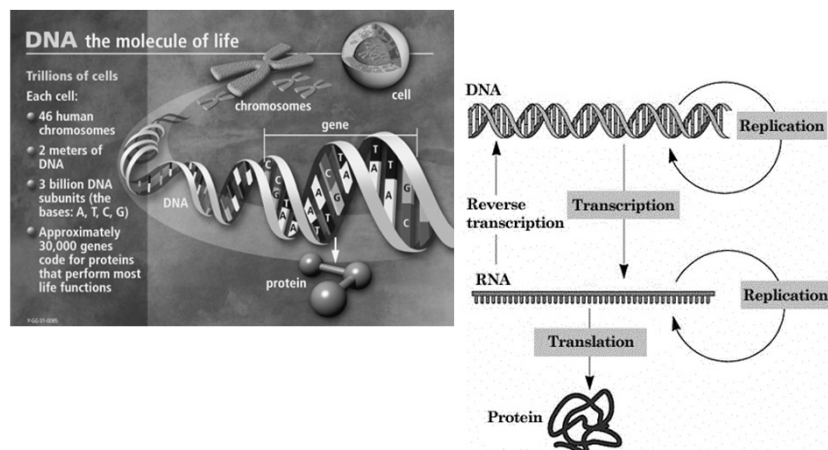
Enzymes

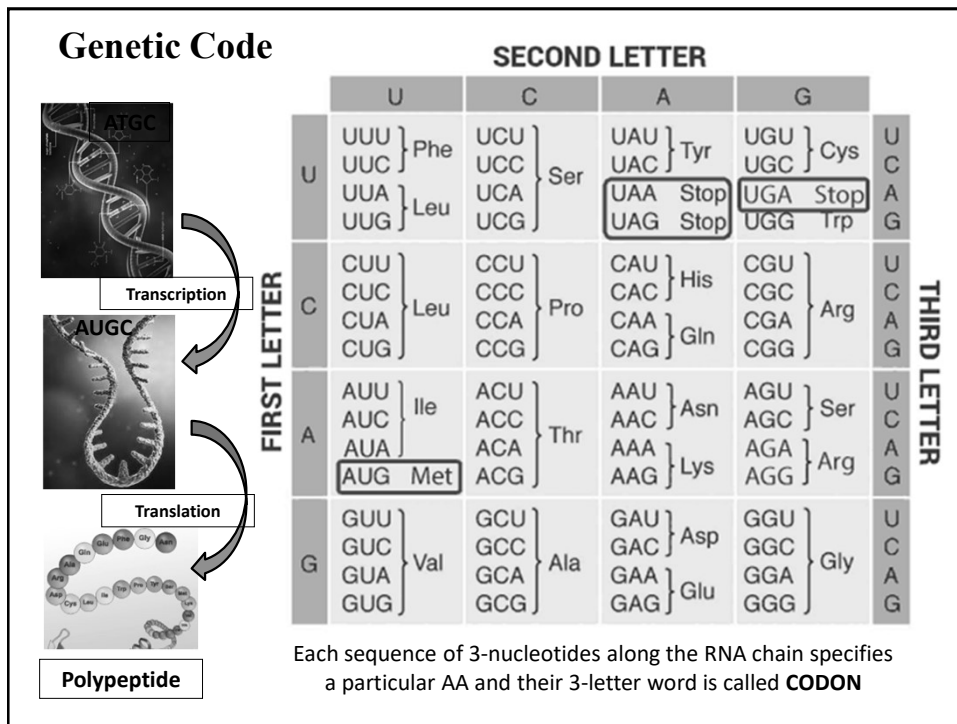
Enzymes are proteins specialized to catalyze biological reactions. They are among the most remarkable biomolecules because of their extraordinary **specificity** and **catalytic power**.

Catalyst

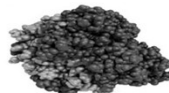
A substance, usually used in small amounts relative to the reactants, that modifies and increases the rate of a reaction without being consumed in the process.

Central Dogma of Molecular Biology



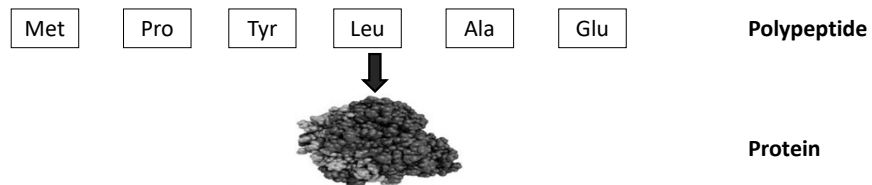


Enzymes

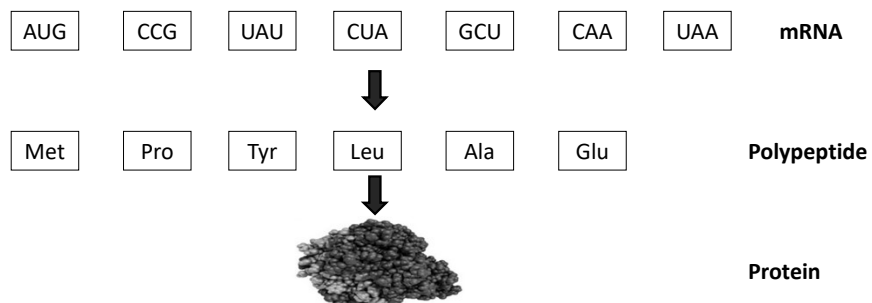


Protein

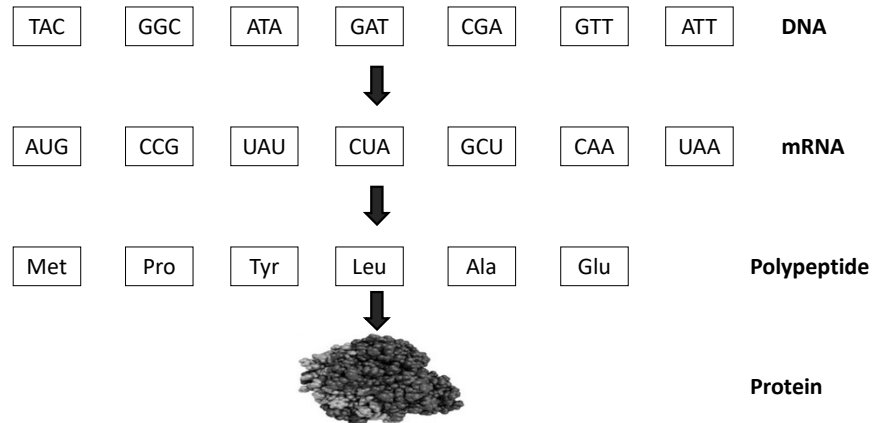
Enzymes



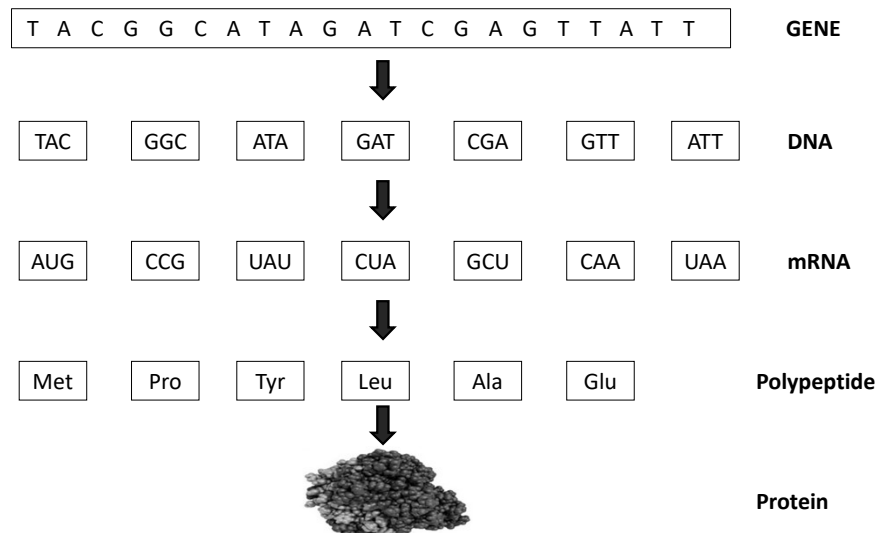
Enzymes



Enzymes



Enzymes



Classification of Enzymes

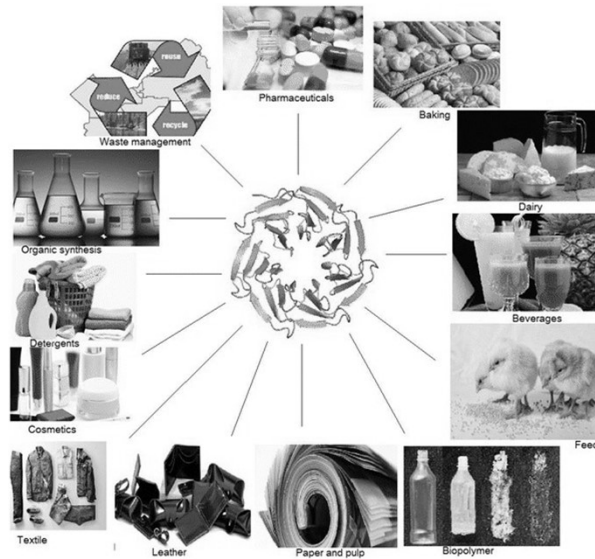
ENZYME CLASS	CHARACTERISTICS	EXAMPLES	REACTION CATALYZED
OXIDOREDUCTASES	Oxidation – Reduction	Alcohol dehydrogenase Peroxidase	$AH_2 + B \longrightarrow A + BH_2$ $Alcohol + NAD^+ \longrightarrow$ $Aldehyde + NADH + H^+$
TRANSFERASES	Transfer of group from one substrate to another other than hydrogen	Hexokinase, Transaminase	$A - X + B \longrightarrow A + B - X$ $Glucose + ATP \longrightarrow Glucose-6-phosphate + ADP + Pi$
HYDROLASES	Cleaves bond and adds water	Choline Esterase, Lipase	$A - B + H_2O \longrightarrow AH + BOH$ $Acetylcholine + H_2O \longrightarrow$ $Acetate + Choline$
LYASES	Addition or removal of a group	Aldolase, Fumarase	$A - B + X - Y \longrightarrow AX - BY$ $Fructose-1,6-biphosphate \longrightarrow$ $Glyceraldehyde-3-phosphate +$ $dihydroxyacetone phosphate$
ISOMERASES	Interconversion of isomers	Triose phosphate isomerase	$A \longrightarrow A'$ $Glyceraldehyde-3-phosphate \longrightarrow$ $dihydroxyacetone phosphate$
LIGASES	Condensation (usually dependent on ATP)	Acetyl CoA Carboxylase, Glutamine synthase	$A + B + ATP \longrightarrow A - B + ADP + Pi$ $Acetyl CoA + ATP \longrightarrow$ $Malonyl CoA + ADP + Pi$

What Do Enzymes Do for You?

SECTOR	APPLICATION AREA	BENEFITS
• Detergents	Household washing and cleaning agents	Wash your clothes in cold water; make your teeth cleaner
• Textiles	Denim washing, silk polishing, leather goods softening	Stonewash your jeans; make cotton look and feel like silk; make your leather soft
• Food processing	Baking, brewing, fruit juice processing	Clarify your juice and beer; make bread better; turn corn starch into sugar syrup
• Pulp and paper	Starch conversion, pitch control, bleach-boosting, deinking, stickies control, slime control	Reduce production costs and improve quality

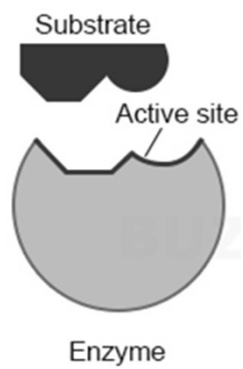
Protease, Lipase, Amylase, Cellulase, Chitinase, Phosphatase.

Enzyme washing or "bio stoning" is a textile manufacturing technique in which cellulase enzymes are applied to denim. As with traditional stone washing, the intended effect is a faded appearance and softer feel.

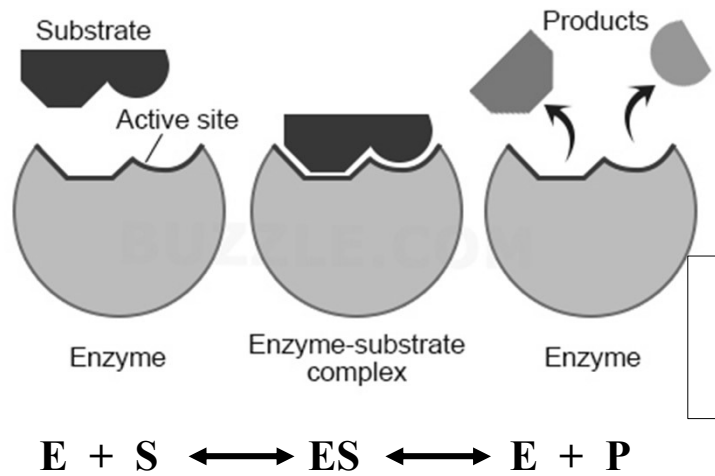


Protease, Lipase, Amylase, Cellulase, Chitinase, Phosphatase.

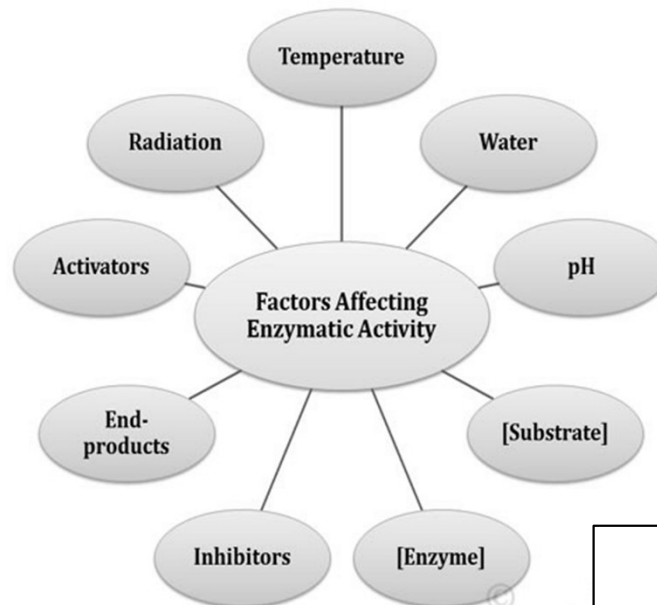
Mechanism of Enzyme action

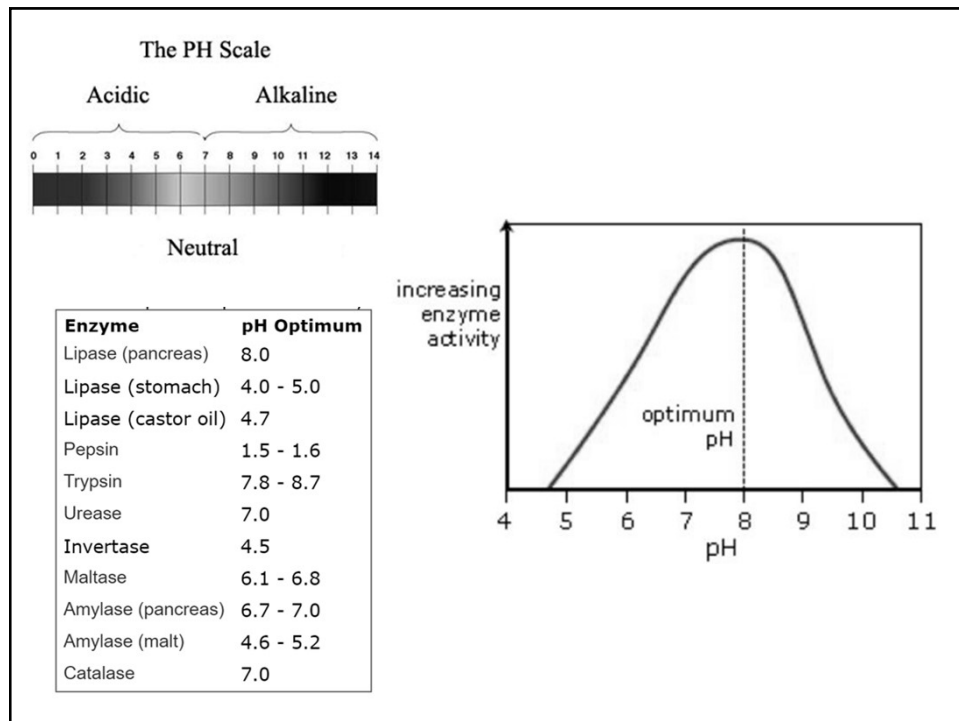
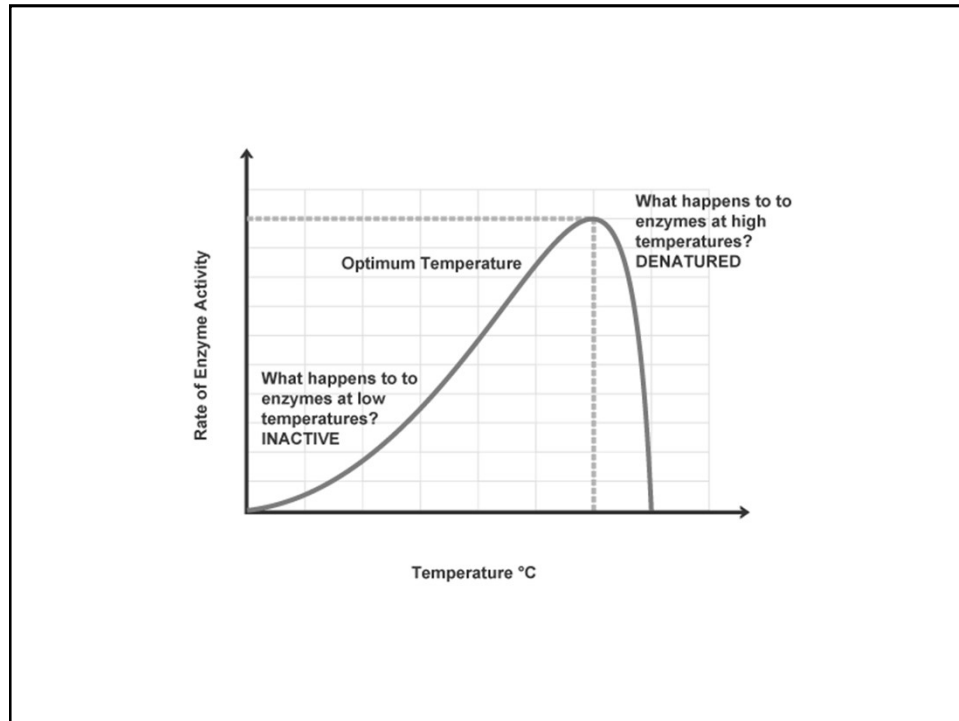


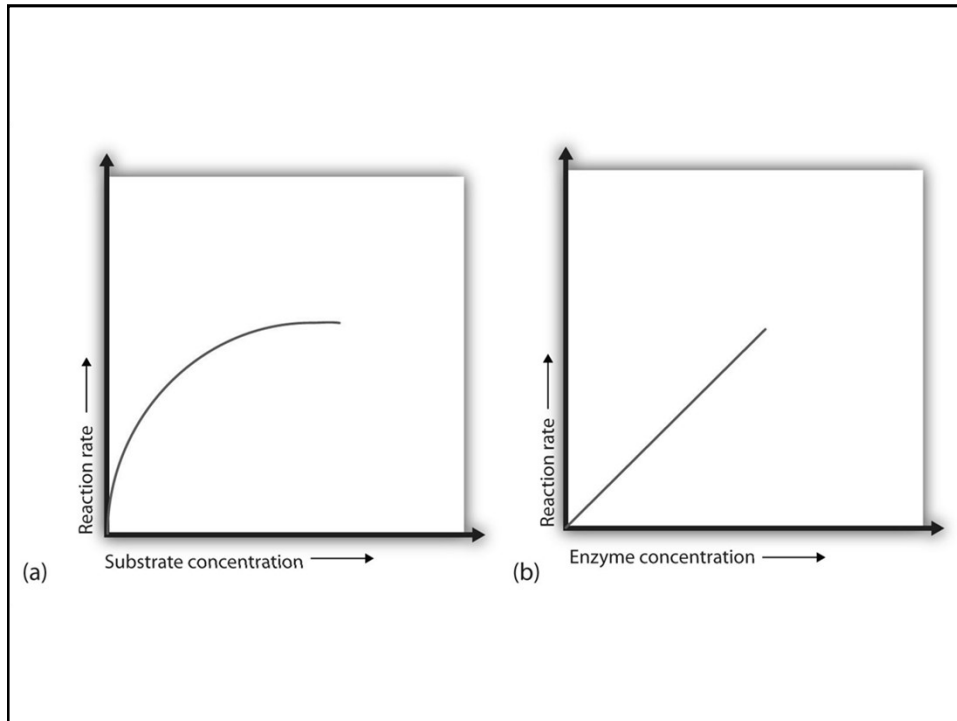
Mechanism of Enzyme action



Factors Affecting Enzymatic Activity

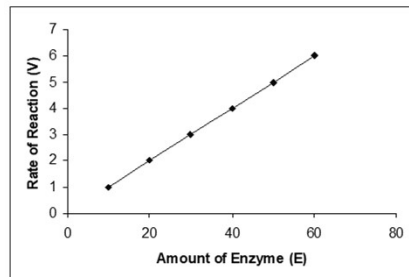






Michaelis-Menten Equation

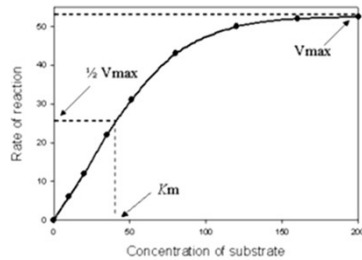
It is one of the simplest and best known models of enzyme kinetics. It describes the rate of enzymatic reaction by relating reaction rate (V) to substrate concentration (S).



Substrate is in excess



Michaelis-Menten Equation



Fixed enzyme concentration

- a = First order kinetics – Phase I
- b = Mixture of zero & I order kinetics
- c = Zero order kinetics – Phase II

$$v = \frac{V_{\max} [S]}{K_M + [S]}$$

V = Observed velocity at a given substrate (S)
 K_m = Michaelis – Menten constant expressed in units of concentration (Moles/l)
 $V_{\max}$ = Velocity at substrate saturation

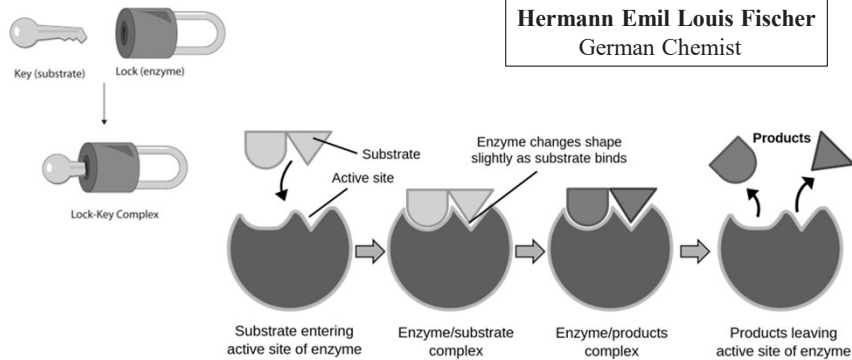


Lock and Key Model

- Emil Fisher 1894
- Both E & S posses specific complimentary geometric shapes



Hermann Emil Louis Fischer
German Chemist

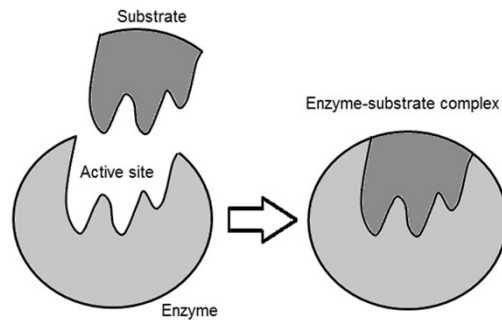


Induced Fit Model

- D. Koshland 1958
- Enzymes are rather flexible structures, the active site is continuously reshaped by interactions with the substrate as the S interacts with the E.

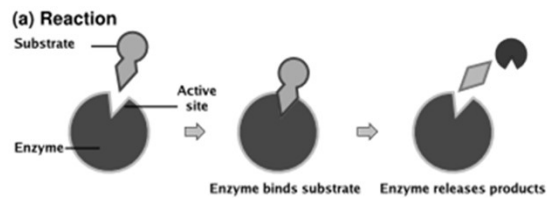


Daniel Koshland
American Biochemist



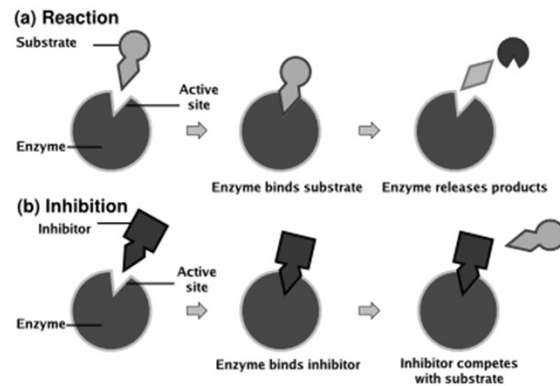
Enzyme Inhibitors

- Compounds that bind with the enzyme in the similar fashion as the substrate binds, and decreases enzyme activity.

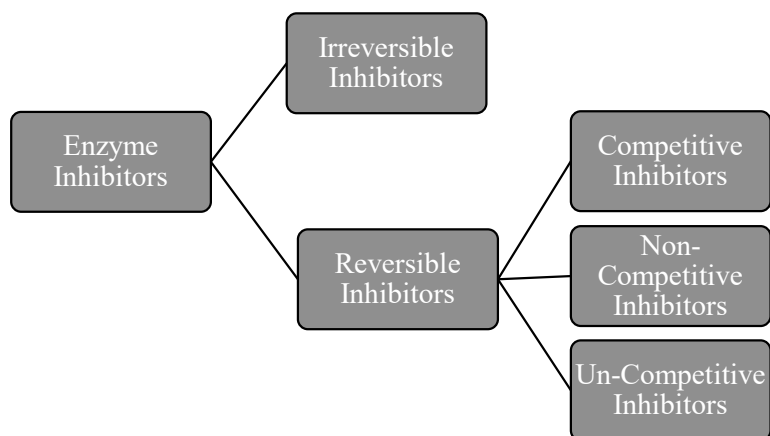


Enzyme Inhibitors

- Compounds that binds with the enzyme in the similar fashion as the substrate binds, and decreases enzyme activity.

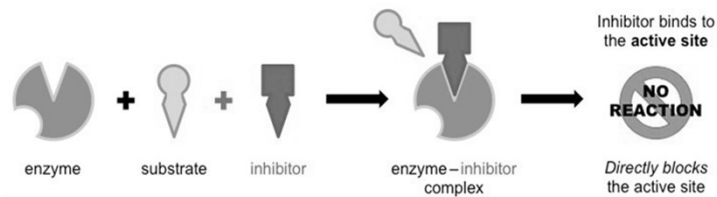
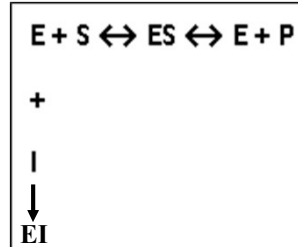


Types of Inhibitors



1. Irreversible Inhibitors

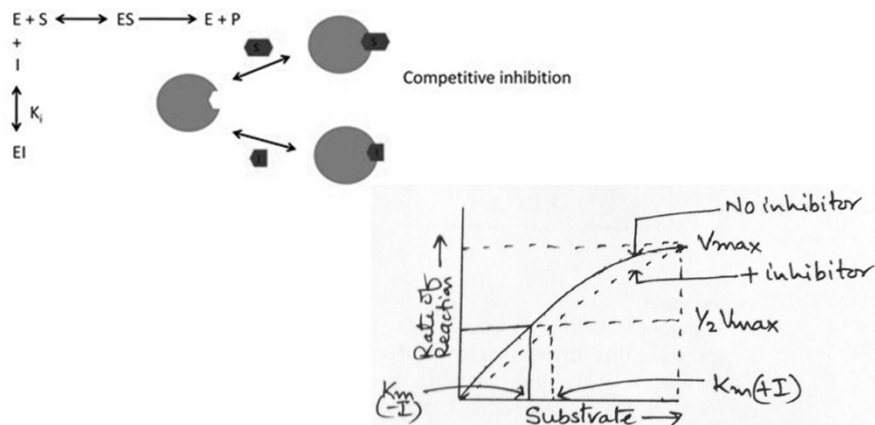
Irreversible inhibitors react with the enzyme and form a covalent adduct with the protein. The inactivation is irreversible. These compounds include **eflornithine** a drug used to treat the parasitic disease sleeping sickness. **Penicillin** and **Aspirin** also act in this manner.



2. Reversible Inhibitors

a). Competitive inhibition

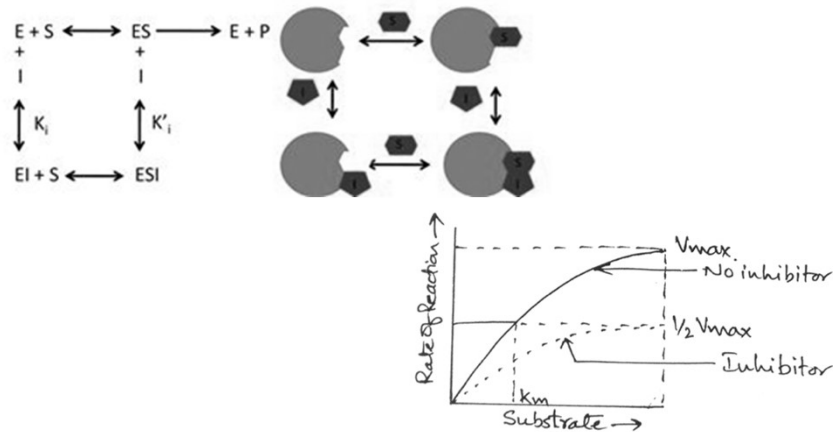
The inhibitor and substrate compete for the enzyme (i.e., they can not bind at the same time). Often competitive inhibitors strongly resemble the real substrate of the enzyme. ES and EI products are formed but EIS products are never produced. And this condition can be overcome by addition of more substrate.



2. Reversible Inhibitors

b). Non-Competitive inhibition

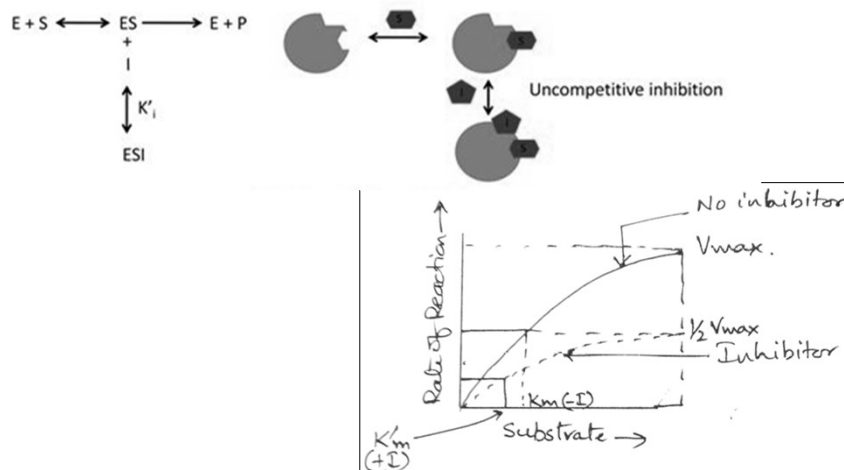
Inhibitors can bind to the enzyme at the binding site at the same time as the substrate. Both the EI and EIS complexes are formed. Because the inhibitor can not be driven from the enzyme by higher substrate concentration (in contrast to competitive inhibition), the apparent $V_{\max}$ changes. But because the substrate can still bind to the enzyme, the K_m stays the same.



2. Reversible Inhibitors

c). Un-Competitive inhibition

In uncompetitive inhibition, the inhibitor cannot bind to the free enzyme, only to the ES-complex. The EIS-complex thus formed is enzymatically inactive. In this case both $V_{\max}$ and K_m will decrease



Applications of Inhibitors

Since inhibitors modulate the function of enzymes they are often used as drugs. A common example of an inhibitor that is used as a drug is aspirin, which inhibits the COX-1 and COX-2 enzymes that produce the inflammation messenger prostaglandin, thus suppressing pain and inflammation.

Therapeutic use of enzyme Inhibitors

Type of enzyme inhibitor	Enzyme inhibitor (drug)	Enzyme Target	Pharmaceutical use
Irreversible inhibitors	Aspirin	Cyclooxygenase	Inflammation/pain/fever
	Penicillin	Transpeptidase	Bacterial infection
	Acyclovir	Viral DNA polymerase	Herpes
	Allopurinol	Xanthine oxidase	Gout
	Amoxicillin	Penicillin binding proteins	Bacterial infection
Competitive reversible inhibitors	Lovastatin, Pravastatin, other statins	HMG-CoA reductase	Cholesterol lowering
	Captopril, enalapril	Angiotensin converting enzyme	Hypertension
	Saquinavir, indinavir, ritonavir	HIV protease	AIDS
	Acetazolamide	Carbonic anhydrase	Glaucoma
	Viagra, Levitra	Phosphodiesterase	Erectile dysfunction
	Gleevec	Bcr-Abl kinase	Cancer
	Methotrexate	Dihydrofolate reductase	Cancer, bacterial infection

DOI: [10.15406/atboa.2019.05.00104](https://doi.org/10.15406/atboa.2019.05.00104)

Therapeutic use of enzyme Inhibitors

Type of enzyme inhibitor	Enzyme inhibitor (drug)	Enzyme Target	Pharmaceutical use
Non-competitive reversible inhibitors	Nevirapine, efavirenz	HIV reverse transcriptase	AIDS
	SB-715992	KSP kinesin	Cancer
	PD0325901, CI-1040	MAP kinase kinase (MEK)	Cancer
	TF-505	Steroid 5 α -reductase	Benign prostate Hyperplasia
	Etoposide	Topoisomerase II	Cancer
	Tacrine	Acetylcholinesterase	Alzheimer's disease
	Trazodone	Adenosine deaminase	Depression
Uncompetitive reversible inhibitors	Mycophenolic acid, VX-148	Inosine 5'-monophosphate dehydrogenase	Immunosuppression, cancer
	Finasteride, epristeride, dutasteride	Steroid 5 α -reductase	Benign prostate hyperplasia, male pattern baldness
	Valproic acid	UDP-glucuronosyltransferases	Xenobiotic metabolism
	Camptothecin	Topoisomerase I	Cancer
	Ciglitazone	15-Hydroxyprostaglandin Dehydrogenase	Inflammatory diseases

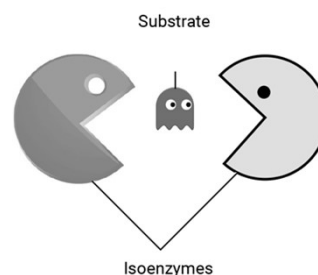
DOI: [10.15406/atropa.2019.05.00104](https://doi.org/10.15406/atropa.2019.05.00104)

Isoenzymes

An enzyme that has multiple molecular forms in the same organism and catalyze the same enzymatic reactions.

Lactic dehydrogenase

- H-LDH
- M-LDH



Allosteric Enzymes

In a multienzyme sequence of reaction some times the end product may act as specific inhibitor for an enzyme at or near the beginning of the sequence. And as a result the rate of entire sequence is determined by the steady state concentration of the end product. This type of inhibition is called the **end product inhibition**, **feed back inhibition** or **retro-inhibition** and the enzyme which is inhibited by the end product is called **allosteric enzyme**.

THANK YOU